



COURSE CODE/TITLE:	CPE212/Automating Server Management	SECTION:	CPE31S2
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ACTIVITY NO. 7.1 Managing Files and Creating Roles in Ansible

1. PROCEDURE OUTPUT

Task 1: Create a file and copy it to remote servers

1. Using the previous directory was created in your previous laboratory, create a directory, and name it "files." Create a file inside that directory and name it "default_site.html." Edit the file and put basic HTML syntax. Any content will do, as long as it will display text later. Save the file and exit.

```
GNU nano 7.2 files/default_site.html *
<!DOCTYPE html>
<html>
<head>
<title>My Web Server</title>
</head>
<body>

<h1>Welcome to My Web Server</h1>

<p>This webpage was deployed using Ansible.</p>

<p>This file was copied from the local repository to the remote server.</p>

</body>
</html>
```

2. Edit the site.yml file and just below the web_servers play, create a new file to copy the default html file for Site

```
- name: copy default html file for site
  tags:
    - apache
    - apache2
    - httpd
  copy:
    src: files/default_site.html
    dest: /var/www/html/index.html
    owner: root
    group: root
    mode: '0644'
```

3. Run the playbook site.yml. Describe the changes.

```
PLAY [web_servers] *****

TASK [Gathering Facts] *****
ok: [192.168.56.101]
ok: [192.168.56.102]
fatal: [192.168.1.23]: FAILED! => {"msg": "Incorrect sudo password"}

TASK [install apache and php for Ubuntu servers] *****
ok: [192.168.56.101]
ok: [192.168.56.102]

TASK [install apache and php for CentOS servers] *****
skipping: [192.168.56.101]
skipping: [192.168.56.102]

TASK [start httpd (CentOS)] *****
skipping: [192.168.56.101]
skipping: [192.168.56.102]

TASK [copy default html file for site] *****
ok: [192.168.56.101]
ok: [192.168.56.102]

PLAY [db_servers] *****

TASK [Gathering Facts] *****
ok: [192.168.56.102]

PLAY [file_servers] *****

TASK [Gathering Facts] *****
ok: [192.168.56.101]

PLAY RECAP *****
192.168.1.23      : ok=0    changed=0    unreachable=0    failed=1    skipped=0    rescued=0    ignored=0
192.168.56.101   : ok=4    changed=0    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0
192.168.56.102   : ok=4    changed=0    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0
```

```

hdjavillonar@workstation:~/CPE232_Javillonar$ ansible-playbook -i inventory --ask-become-pass site.yml --tags apache
BECOME password:

PLAY [web_servers] *****

TASK [Gathering Facts] *****
ok: [192.168.1.23]
fatal: [192.168.56.101]: FAILED! => {"msg": "Incorrect sudo password"}
fatal: [192.168.56.102]: FAILED! => {"msg": "Incorrect sudo password"}

TASK [install apache and php for Ubuntu servers] *****
skipping: [192.168.1.23]

TASK [install apache and php for CentOS servers] *****
ok: [192.168.1.23]

TASK [start httpd (CentOS)] *****
ok: [192.168.1.23]

TASK [copy default html file for site] *****
changed: [192.168.1.23]

PLAY [db_servers] *****

TASK [Gathering Facts] *****
ok: [192.168.1.23]

PLAY [file_servers] *****

PLAY RECAP *****
192.168.1.23      : ok=5    changed=1    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0
192.168.56.101   : ok=0    changed=0    unreachable=0    failed=1    skipped=0    rescued=0    ignored=0
192.168.56.102   : ok=0    changed=0    unreachable=0    failed=1    skipped=0    rescued=0    ignored=0

```

- The Apache and PHP packages were successfully installed, and the HTML file was successfully copied to both Ubuntu web servers (192.168.56.101 and 192.168.56.102). However, the CentOS server (192.168.1.23) failed due to an incorrect sudo password.

4. Go to the remote servers (web_servers) listed in your inventory. Use cat command to check if the index.html is the same as the local repository file (default_site.html). Do both for Ubuntu and CentOS servers. On the CentOS server, go to the browser and type its IP address.

FROM SERVER 1:

```
wdjavillonar@workstation:~/CPE232_Javillonar$ ssh serv1javillonar@192.168.56.101
Welcome to Ubuntu 24.04.5 LTS (GNU/Linux 7.0.0-30-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

*** System restart required ***
Last login: Thu Sep 10 11:52:23 2026 from 192.168.56.103
serv1javillonar@serv1javillonar-VirtualBox:~$ cat /var/www/html/index.html
<!DOCTYPE html>
<html>
<head>
<title>My Web Server</title>
</head>
<body>

<h1>Welcome to My Web Server</h1>

<p>This webpage was deployed using Ansible.</p>

<p>This file was copied from the local repository to the remote server
.</p>

</body>
</html>
```

FROM S2

```
Connection to 192.168.56.101 closed.
wdjavillonar@workstation:~/CPE232_Javillonar$ ssh serv2javillonar@192.168.56.102
Welcome to Ubuntu 24.04.5 LTS (GNU/Linux 7.0.0-30-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

*** System restart required ***
Last login: Thu Sep 10 11:52:23 2026 from 192.168.56.103
serv2javillonar@server2:~$ cat /var/www/html/index.html
<!DOCTYPE html>
<html>
<head>
<title>My Web Server</title>
</head>
<body>

<h1>Welcome to My Web Server</h1>

<p>This webpage was deployed using Ansible.</p>

<p>This file was copied from the local repository to the remote server
.</p>

</body>
</html>
```

FROM CENTOS

```
serv2javillonar@server2:~$ ssh cjavillonar@192.168.1.23
The authenticity of host '192.168.1.23 (192.168.1.23)' can't be established.
ED25519 key fingerprint is SHA256:+a6Wj1Yfn4sLSn6KfvIR4Zu2gb6Qxqh4IDM1
4376pLM.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? y
es
Warning: Permanently added '192.168.1.23' (ED25519) to the list of known hosts.
cjavillonar@192.168.1.23's password:
Activate the web console with: systemctl enable --now cockpit.socket

Last login: Thu Sep 10 11:52:33 2026 from 192.168.1.171
[cjavillonar@localhost ~]$ cat /var/www/html/index.html
<!DOCTYPE html>
<html>
<head>
<title>My Web Server</title>
</head>
<body>

<h1>Welcome to My Web Server</h1>

<p>This webpage was deployed using Ansible.</p>

<p>This file was copied from the local repository to the remote server
.</p>

</body>
</html>
```

Describe the output.

The original tasks were reorganized into separate Ansible roles for the base system, workstations, web servers, database servers, and file servers. The main site.yml now calls these roles, making the playbook more organized and easier to manage.

5. Sync your local repository with GitHub and describe the changes.

```
wdjavillonar@workstation:~/CPE232_Javillonar$ git status
On branch main
Your branch is up to date with 'origin/main'.

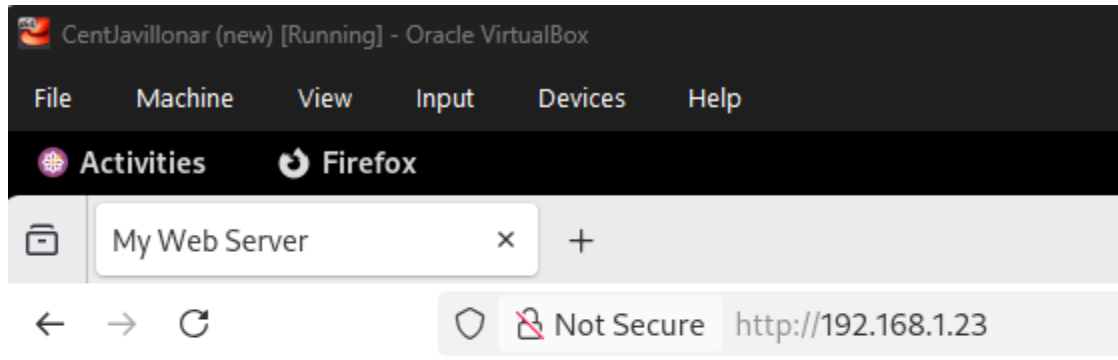
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
)
        modified:   install_apache.yml

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        files/
        inventory
        inventory.yml
        redhat_apache.yml
        roles.yml
        roles/
        site.yml

no changes added to commit (use "git add" and/or "git commit -a")
wdjavillonar@workstation:~/CPE232_Javillonar$ git add .
wdjavillonar@workstation:~/CPE232_Javillonar$ git commit -m "Complete Task 1"
[main 19624c0] Complete Task 1
 9 files changed, 218 insertions(+), 9 deletions(-)
 create mode 100644 files/default_site.html
 create mode 100644 inventory
 create mode 100644 inventory.yml
 create mode 100644 redhat_apache.yml
 create mode 100644 roles.yml
 create mode 100644 roles/db/tasks/main.yml
 create mode 100644 roles/web/tasks/main.yml
 create mode 100644 site.yml

wdjavillonar@workstation:~/CPE232_Javillonar$ git status
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
```



Welcome to My Web Server

This webpage was deployed using Ansible.

This file was copied from the local repository to the remote server.

Task 2: Download a file and extract it to a remote server

1. Edit the site.yml. Just before the web_servers play, create a new play

```
- name: install updates (Ubuntu)
  apt:
    upgrade: dist
    update_cache: yes
  when: ansible_distribution == "Ubuntu"

- hosts: workstations
  become: true

  tasks:

    - name: install unzip
      package:
        name: unzip

    - name: install terraform
      unarchive:
        src: https://releases.hashicorp.com/terraform/0.12.28/terraform_0.12.28_Linux_x86_64.tar.gz
        dest: /usr/local/bin
        remote_src: yes
        mode: 0755
        owner: root
        group: root

- hosts: web_servers
  become: true

  tasks:

    - name: install apache and php for Ubuntu servers
      tags:
        - apache
```

^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute
 ^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify

2. Edit the inventory file and add a workstation group. Add any Ubuntu remote server. Make sure to remember the IP address.

```
- hosts: file_servers
  become: true

  tasks:

    - name: install samba package
      tags:
        - samba
      package:
        name: samba
        state: latest

- hosts: workstations
  become: true

  tasks:

    - name: install unzip
      package:
        name: unzip
        state: present

    - name: install terraform
      unarchive:
        src: https://releases.hashicorp.com/terraform/0.12.28/terraform_0.12.28_linux_amd64.zip
        dest: /usr/local/bin
        remote_src: yes
        mode: '0755'
        owner: root
        group: root
```

^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute
 ^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify

3. Run the playbook.

```
wdjavillonar@workstation:~/CPE232_Javillonar$ ansible-playbook -i inventory --ask-pass --ask-become-pass site.yml --limit workstations --start-at-task "install unzip"
SSH password:
BECOME password[defaults to SSH password]:

PLAY [all] *****
*****

PLAY [workstations] *****
*****

TASK [Gathering Facts] *****
*****
ok: [192.168.56.103]

TASK [install unzip] *****
*****
ok: [192.168.56.103]

TASK [install terraform] *****
*****
changed: [192.168.56.103]

PLAY RECAP *****
192.168.56.103      : ok=3    changed=1    unreachable=0    failed=0
                    skipped=0    rescued=0    ignored=0

wdjavillonar@workstation:~/CPE232_Javillonar$ S
```

Describe the output.

4. On the Ubuntu remote workstation, type terraform to verify installation of terraform. Describe the output

```
wdjavillonar@workstation:~/CPE232_Javillonar$ terraform
Usage: terraform [-version] [-help] <command> [args]

The available commands for execution are listed below.
The most common, useful commands are shown first, followed by
less common or more advanced commands. If you're just getting
started with Terraform, stick with the common commands. For the
other commands, please read the help and docs before usage.

Common commands:
  apply          Builds or changes infrastructure
  console        Interactive console for Terraform interpolations
  destroy        Destroy Terraform-managed infrastructure
  env            Workspace management
  fmt            Rewrites config files to canonical format
  get            Download and install modules for the configuration
  graph          Create a visual graph of Terraform resources
  import         Import existing infrastructure into Terraform
  init           Initialize a Terraform working directory
  login          Obtain and save credentials for a remote host
  logout         Remove locally-stored credentials for a remote host
  output         Read an output from a state file
  plan           Generate and show an execution plan
  providers      Prints a tree of the providers used in the configuration
  refresh        Update local state file against real resources
  show           Inspect Terraform state or plan
  taint          Manually mark a resource for recreation
  untaint        Manually unmark a resource as tainted
  validate       Validates the Terraform files
  version        Prints the Terraform version
  workspace      Workspace management

All other commands:
  0.12upgrade    Rewrites pre-0.12 module source code for v0.12
  debug          Debug output management (experimental)
  force-unlock   Manually unlock the terraform state
  push           Obsolete command for Terraform Enterprise legacy (v1)
  state          Advanced state management
```

Task 3: Create roles

1. Edit the site.yml. Configure roles as follows: (make sure to create a copy of the old site.yml file because you will be copying the specific plays for all groups)

```
wdjavillonar@workstation:~/CPE232_Javillonar$
wdjavillonar@workstation:~/CPE232_Javillonar$ cp site.yml site_task2_b
ackup.yml
wdjavillonar@workstation:~/CPE232_Javillonar$ ls
files                README.md            site_task2_backup.yml
install_apache.yml  redhat_apache.yml  site.yml
inventory           roles
inventory.yml       roles.yml
wdjavillonar@workstation:~/CPE232_Javillonar$ S
```

```
---
- hosts: all
  become: true
  pre_tasks:
    - name: update repository index (CentOS)
      tags: always
      dnf:
        update_cache: yes
        changed_when: false
        when: ansible_distribution == "CentOS"
    - name: install updates (Ubuntu)
      tags: always
      apt:
        update_cache: yes
        changed_when: false
        when: ansible_distribution == "Ubuntu"

- hosts: all
  become: true
  roles:
    - base

- hosts: workstations
  become: true
  roles:
    - workstations

- hosts: web_servers
  become: true
  roles:
    - web_servers

- hosts: db_servers
  become: true
  roles:
    - db_servers

- hosts: file_servers
  become: true
  roles:
    - file_servers
```

```
GNU nano 7.2                                site.yml *
---
- hosts: all
  become: true

  pre_tasks:

    - name: update repository index (CentOS)
      tags: always
      dnf:
        update_cache: yes
        changed_when: false
        when: ansible_distribution == "CentOS"

    - name: install updates (Ubuntu)
      tags: always
      apt:
        update_cache: yes
        changed_when: false
        when: ansible_distribution == "Ubuntu"

- hosts: all
  become: true
  roles:
    - base

- hosts: workstations
  become: true
  roles:
    - workstations

- hosts: web_servers
  become: true
  roles:
    - web_servers
```

2. Under the same directory, create a new directory and name it roles. Enter the roles directory and create new directories: base, web_servers, file_servers, db_servers and workstations. For each directory, create a directory and name it tasks.

```
wdjavillonar@workstation:~/CPE232_Javillonar$ nano site.yml
wdjavillonar@workstation:~/CPE232_Javillonar$ mkdir -p roles/base/tasks
wdjavillonar@workstation:~/CPE232_Javillonar$ mkdir -p roles/web_servers/tasks
wdjavillonar@workstation:~/CPE232_Javillonar$ mkdir -p roles/web_servers/files
wdjavillonar@workstation:~/CPE232_Javillonar$ mkdir -p roles/file_servers/tasks
wdjavillonar@workstation:~/CPE232_Javillonar$ mkdir -p roles/db_servers/tasks
wdjavillonar@workstation:~/CPE232_Javillonar$ mkdir -p roles/workstations/tasks
wdjavillonar@workstation:~/CPE232_Javillonar$ find roles -type d
roles
roles/web_servers
roles/web_servers/files
roles/web_servers/tasks
roles/file_servers
roles/file_servers/tasks
roles/db
roles/db/tasks
roles/web
roles/web/tasks
roles/db_servers
roles/db_servers/tasks
roles/base
roles/base/tasks
roles/workstations
roles/workstations/tasks
```

3. Go to tasks for all directory and create a file. Name it main.yml. In each of the tasks for all directories, copy and paste the code from the old site.yml file. Show all contents of main.yml files for all tasks.

```
GNU nano 7.2 roles/base/tasks/main.yml *
---
- name: install updates (CentOS)
  dnf:
    update_only: yes
    update_cache: yes
  when: ansible_distribution == "CentOS"

- name: install updates (Ubuntu)
  apt:
    upgrade: dist
    update_cache: yes
  when: ansible_distribution == "Ubuntu"
```

```
GNU nano 7.2      roles/web_servers/tasks/main.yml *
---
- name: install apache and php for Ubuntu servers
  tags:
    - apache
    - ubuntu
  apt:
    name:
      - apache2
      - libapache2-mod-php
    state: latest
  when: ansible_distribution == "Ubuntu"

- name: install apache and php for CentOS servers
  tags:
    - apache
    - centos
    - httpd
  dnf:
    name:
      - httpd
      - php
    state: latest
  when: ansible_distribution == "CentOS"

- name: start httpd (CentOS)
  tags:
    - apache
    - centos
    - httpd
  service:
    name: httpd
    state: started
    enabled: true
  when: ansible_distribution == "CentOS"
```

GNU nano 7.2 roles/db_servers/tasks/main.yml *

```
---
- name: install mariadb package (CentOS)
  tags:
    - centos
    - db
    - mariadb
  dnf:
    name: mariadb-server
    state: latest
  when: ansible_distribution == "CentOS"

- name: install mariadb package (Ubuntu)
  tags:
    - db
    - mariadb
    - ubuntu
  apt:
    name: mariadb-server
    state: latest
  when: ansible_distribution == "Ubuntu"

- name: MariaDB-Restarting/Enabling
  tags:
    - db
    - mariadb
  service:
    name: mariadb
    state: restarted
    enabled: trueS
```

```
GNU nano 7.2      roles/file_servers/tasks/main.yml *
---
- name: install samba package
  tags:
    - samba
  package:
    name: samba
    state: latesto

GNU nano 7.2      roles/workstations/tasks/main.yml *
---
- name: install unzip
  package:
    name: unzip

- name: install terraform
  unarchive:
    src: https://releases.hashicorp.com/terraform/0.12.28/terraform_0
    dest: /usr/local/bin
    remote_src: yes
    mode: 0755
    owner: root
    group: root

wdjavillonar@workstation:~/CPE232_Javillonar$ find roles -type f
roles/web_servers/files/default_site.html
roles/web_servers/tasks/main.yml
roles/file_servers/tasks/main.yml
roles/db/tasks/main.yml
roles/web/tasks/main.yml
roles/db_servers/tasks/main.yml
roles/base/tasks/main.yml
roles/workstations/tasks/main.yml
wdjavillonar@workstation:~/CPE232_Javillonar$
```

4. Run the site.yml playbook and describe the output.



```
PLAY [all] *****

TASK [Gathering Facts] *****
ok: [192.168.56.103]
ok: [192.168.56.102]
ok: [192.168.56.101]
fatal: [192.168.1.23]: FAILED! => {"msg": "Incorrect sudo password"}

TASK [update repository index (CentOS)] *****
skipping: [192.168.56.103]
skipping: [192.168.56.101]
skipping: [192.168.56.102]

TASK [install updates (Ubuntu)] *****
ok: [192.168.56.103]
ok: [192.168.56.101]
ok: [192.168.56.102]

PLAY [all] *****

TASK [Gathering Facts] *****
ok: [192.168.56.102]
ok: [192.168.56.103]
ok: [192.168.56.101]

TASK [base : install updates (CentOS)] *****
skipping: [192.168.56.103]
skipping: [192.168.56.101]
skipping: [192.168.56.102]

TASK [base : install updates (Ubuntu)] *****
ok: [192.168.56.101]
ok: [192.168.56.103]
ok: [192.168.56.102]

PLAY [workstations] *****

TASK [Gathering Facts] *****
ok: [192.168.56.103]
```

```
TASK [web_servers : install apache and php for Ubuntu servers] *****
ok: [192.168.56.102]
ok: [192.168.56.101]

TASK [web_servers : install apache and php for CentOS servers] *****
skipping: [192.168.56.101]
skipping: [192.168.56.102]

TASK [web_servers : start httpd (CentOS)] *****
skipping: [192.168.56.101]
skipping: [192.168.56.102]

TASK [web_servers : copy default html file for site] *****
ok: [192.168.56.101]
ok: [192.168.56.102]

PLAY [db_servers] *****

TASK [Gathering Facts] *****
ok: [192.168.56.102]

TASK [db_servers : install mariadb package (CentOS)] *****
skipping: [192.168.56.102]

TASK [db_servers : install mariadb package (Ubuntu)] *****
ok: [192.168.56.102]

TASK [db_servers : MariaDB-Restarting/Enabling] *****
changed: [192.168.56.102]

PLAY [file_servers] *****

TASK [Gathering Facts] *****
ok: [192.168.56.101]

TASK [file_servers : install samba package] *****
ok: [192.168.56.101]

PLAY RECAP *****
PLAY RECAP *****
192.168.1.23      : ok=0    changed=0    unreachable=0    failed=1    skipped=0    rescued=0    ignored=0
192.168.56.101   : ok=9    changed=0    unreachable=0    failed=0    skipped=4    rescued=0    ignored=0
192.168.56.102   : ok=10   changed=1    unreachable=0    failed=0    skipped=5    rescued=0    ignored=0
192.168.56.103   : ok=7    changed=0    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0
wdiavillonar@workstation:~/CP5232_davillonar$
```

- The playbook successfully executed the configured roles for the different groups of servers. The base, workstation, web server, database server, and file server roles were applied according to the inventory. The PLAY RECAP shows the status of each managed host, and a successful run should have no failed or unreachable hosts.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Answer the following:

1. What is the importance of creating roles?

- Creating roles is important because it helps organize the Ansible playbook into separate sections based on their purpose. Instead of putting all the tasks in one large site.yml, each role can handle a specific function like web servers, database servers, or workstations. This makes the playbook easier to understand, manage, and modify.

2. What is the importance of managing files?

- Managing files is important because Ansible allows us to automatically copy, create, and configure files on remote machines. This makes the setup faster and more consistent since we don't have to manually configure every server. In our activity, we used file management to deploy the HTML file to the web servers.

Conclusions/Learnings

In this activity, I learned how to organize an Ansible project using roles and how to manage files across different remote machines. I also learned that small errors in the configuration can cause the playbook to fail, so checking the YAML files carefully is important. I encountered the 403 Forbidden error multiple times, especially when updating packages from the Ubuntu repository, which taught me that repository or mirror issues can also affect Ansible tasks. Overall, I learned that Ansible makes server management more organized and automated, but proper configuration and troubleshooting are still necessary.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: Claude

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: Used for helping me remember the commands and fix error 403.

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: AI was used just to assist me and i can confirm that it did not replace my own original thinking and understanding.

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